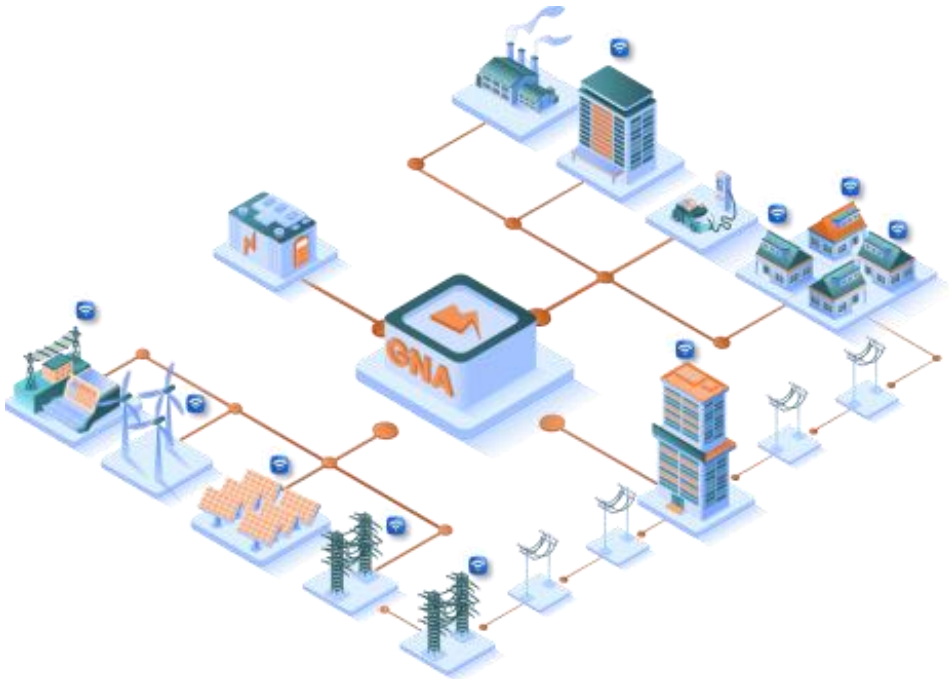
GNA Energy



Presentation for ISUW

ISO/IEC 27001:2022 Certified
ISO 22301:2019 Certified

Regulated by Hon'ble Central Electricity Regulatory Commission

Background on OTC Platforms

Over-the-Counter (OTC) Platforms were approved under Power Market Regulations (PMR) 2021 and are regulated by the Central Electricity Regulatory Commission (CERC)

PMR 2021: Definition and Objectives of an OTC Platform

- To provide an electronic platform for the exchange of information among the potential buyers and sellers of electricity
- To maintain a repository of data related to buyers and sellers and provide such historical data to Market Participants
- To provide such services as advanced data analysis tools to Market Participants

About Us

GNA Energy OTC Platform is a cutting edge platform that revolutionises the way you access and manage your electricity portfolio.

Major States and Generators Onboarded

60+ BUS Listed

5+ BUs Closed

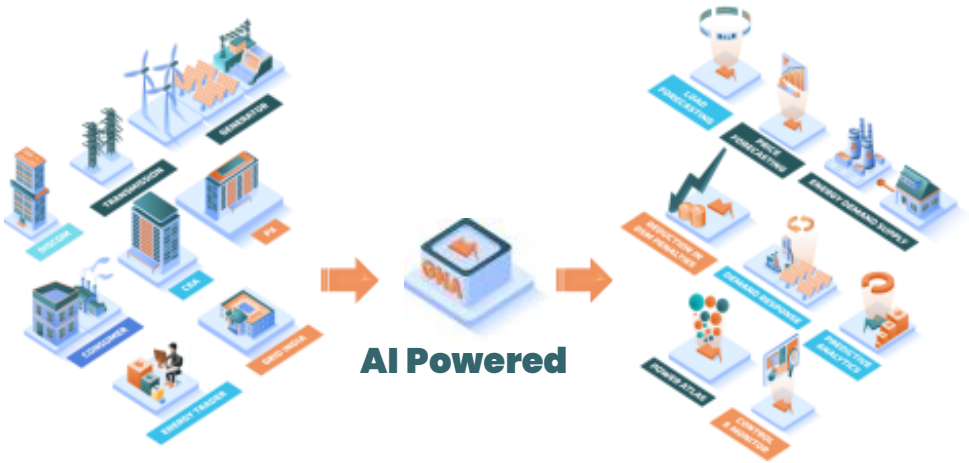
CERC Approved Platform

OTC Platform: 24x7 Digital Market for Electricity for Custom Requirements

GNAi: AI Powered Unified Analytics Platform for the Power Market

Vidyut AI, India's First AI Agent Specifically Built for the Indian Power Market

GNAi – AI Powered Data and Analytics Platform for Power



DATA

INSIGHTS

Centralised Dataset

Easy to Read and Analyse
Automated Reports

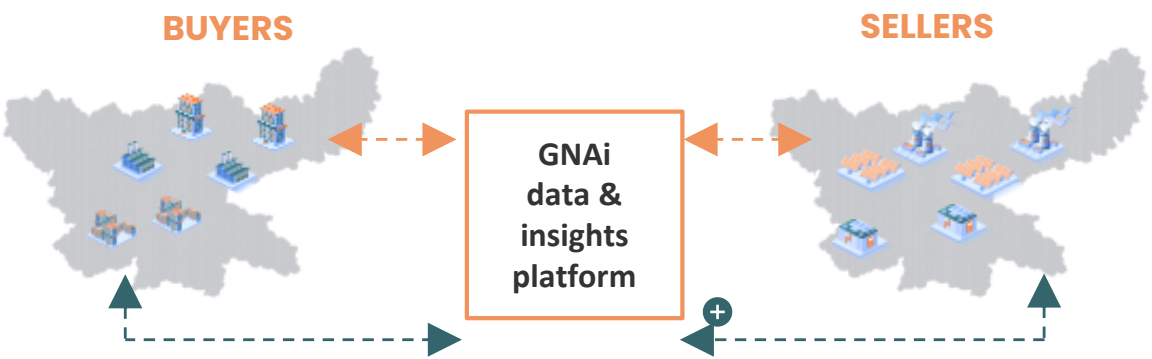
Forecasting

Price Forecasting, Generation & Load Forecasting, Resource Adequacy Planning

Smart Document Repository

Policy Documents, Circulars, Reports, etc.

GNAi – Helping Utilities to Overcome Power Procurement Challenges



- Changing Energy Mix
- Multiple Market Products (DAM/RTM/ GDAM) – Simultaneous Optimization
- Extreme weather variability
- Managing Long Term PPAs vis-à-vis Short-Term procurement
- Managing DSM

GNA Energy's Data Framework

DATA SOURCES

- CEA
- CERC
- State Regulatory Commissions
- GRID India
- NLDC
- RLDCs
- Regional Power Commissions
- DISCOMs
- Generators
- C&I Consumers
- Power Exchanges
- OTC Platforms
- Traders
- 3rd Party Databases
- International Data for Benchmarking

DATA FORMATS

- APIS
- MS-Excel
- CSV
- PDFs
- Scanned PDFs
- Images
- Websites
- Databases
- Internal Systems

DATA PROCESSING

- Automated + Manual Data Sourcing
- Automated Data Validation
- Automated Processing
  - Define Schematic relationship between datasets reported across different sources
  - Standardise reporting formats and units
- Manual Quality Checks

INDIAN POWER MARKET KNOWELDGE HUB

Central Data Repository

Central Document Repository

[Regulatory documents, ARR, Tariff Policy, Newsletters, Reports, Thought Leadership, etc.]

REGULATORS, POLICY MAKERS, & OTC PLATFORM MEMBERS/SUBSCRIBERS

Centralised Data and Document Search Engine

Reporting and Data Visualisation

Advanced Modelling, Machine Learning and AI

Application Areas

Demand Forecasting

Machine Learning Based models

- Historical Patterns
- Weather Variability
- Real Time Data
- Economic Activity
- Festival & Holidays

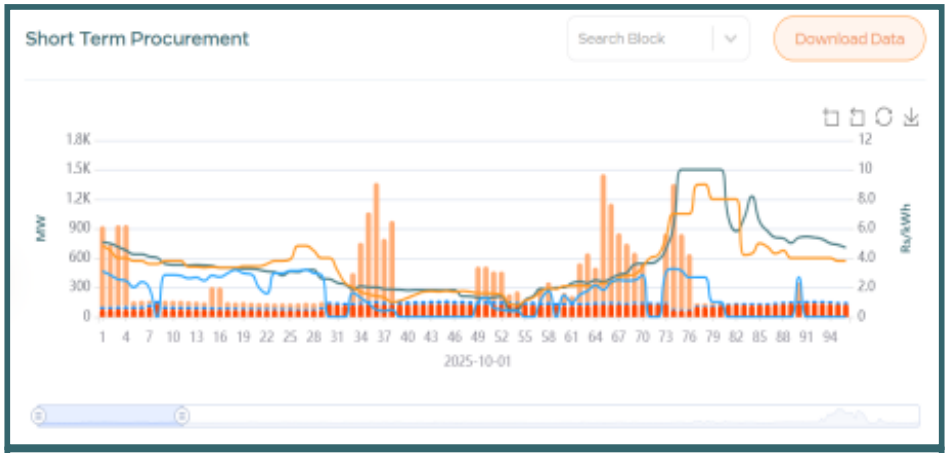
Scheduling Optimization & Unit Commitment

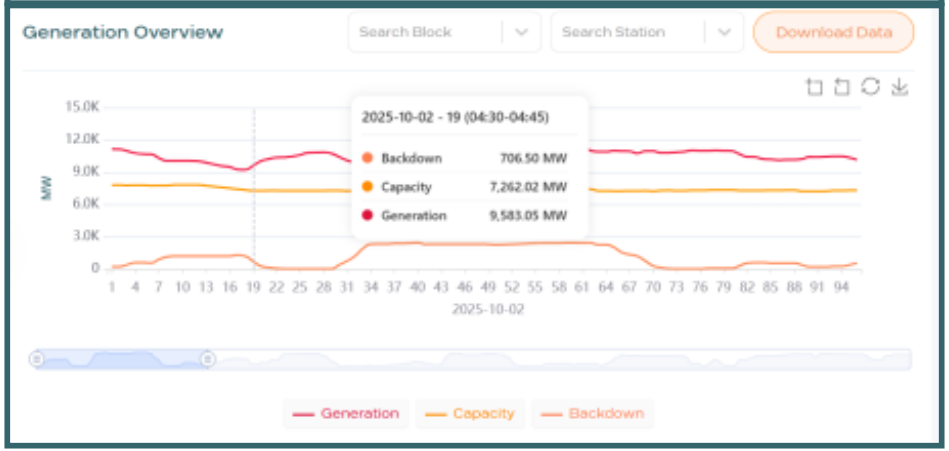
- Ramping Constraints
- Must Run Commitments
- Technical Constraints
- RE Priority

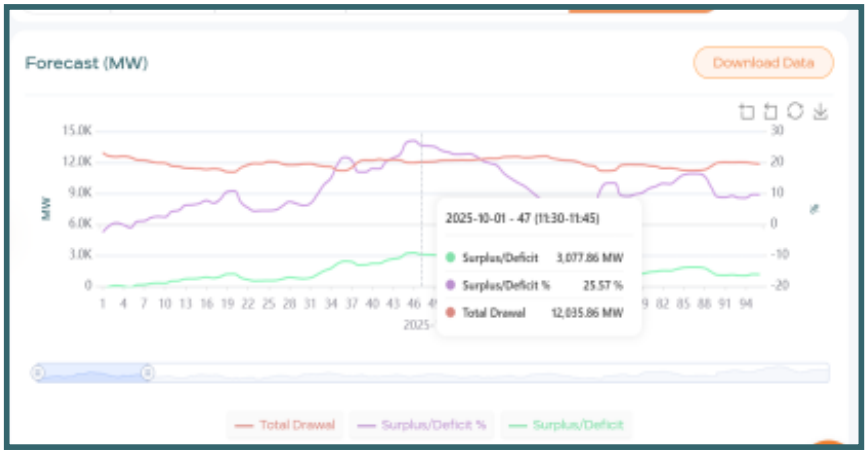
Market Inteligence & Price Forecasting

- MCP Forecasting across time range
- Automated Bidding Management
- Congestion & Corriddor availability

GNAi – SAAS Platform Data Snapshot

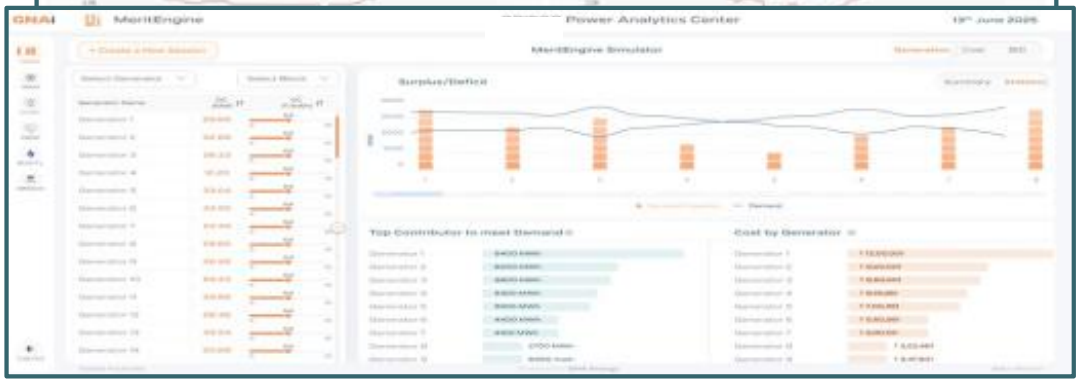






- Regional Virtual Data Hubs for the State utilities
- Substation wise data capture

GNAi – Automated MoD and Simulator Module



GNAi – Automated MoD and Simulator Module

- Manual Process with Possibility of Error
- Generators Clubbed into Broader Groups to Save Time
- No Scenario Testing
- Lack of Transparency & Auditability
- Scalability Challenges with Increase in RE and # of Units

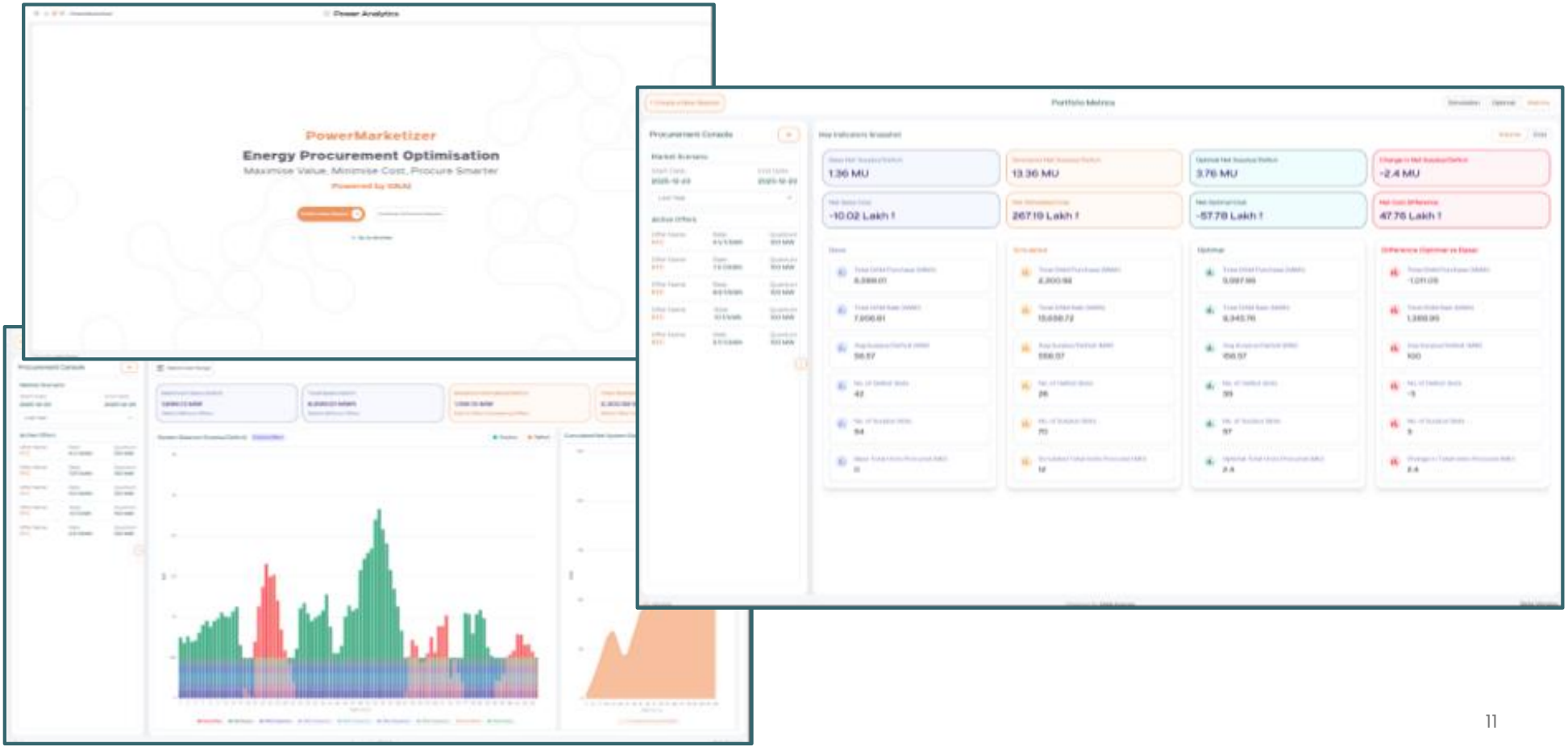
GNAi

Integrated Analytics

- Load Forecast
- Price Forecast
- Surplus/Deficit Profile
- Generator Parameters (VC, DC, Must-run, Tech. Min.)
- Buy/Sell Bid Criteria

- Optimal Dispatch Schedule with Generator Level Calculations
- Automated LGBR Report (Load Generation Balance Report)
- Automated MoD
- Simulation Modules to Stress-test Extreme Events/Deviations
- Pre and Post Bid Financial Analysis
- Standardised Process with Complete Audit Trail

GNAi – Procurement Optimisation



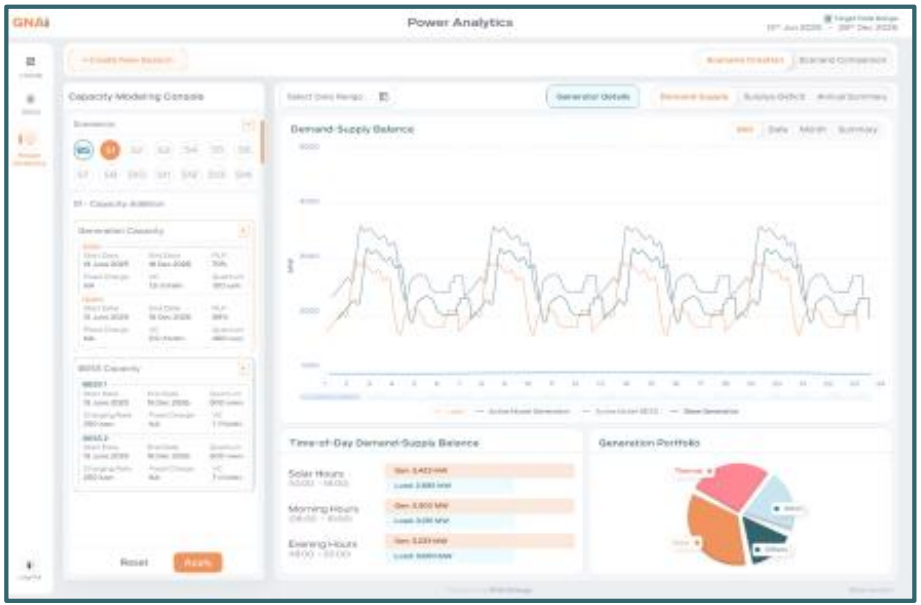
GNAi – Procurement Optimisation

Optimise procurement strategy to minimise procurement costs while reliably meeting power deficit requirements.

1. Assess surplus and shortfall positions across time blocks after accounting for available bilateral and banking arrangements.
2. Analyse the impact of DAM rates on cost and for any remaining surplus or deficit.
3. Identify the most cost-optimal option(s) to efficiently manage surplus and deficit power.
4. Simulate different market scenarios to test strategies under changing price and demand conditions.
5. Assess risk exposure and worst-case outcomes, stress-testing all options to select resilient and cost-efficient procurement strategies.

GNAi – Optimisation & Simulation – Long –term Planning

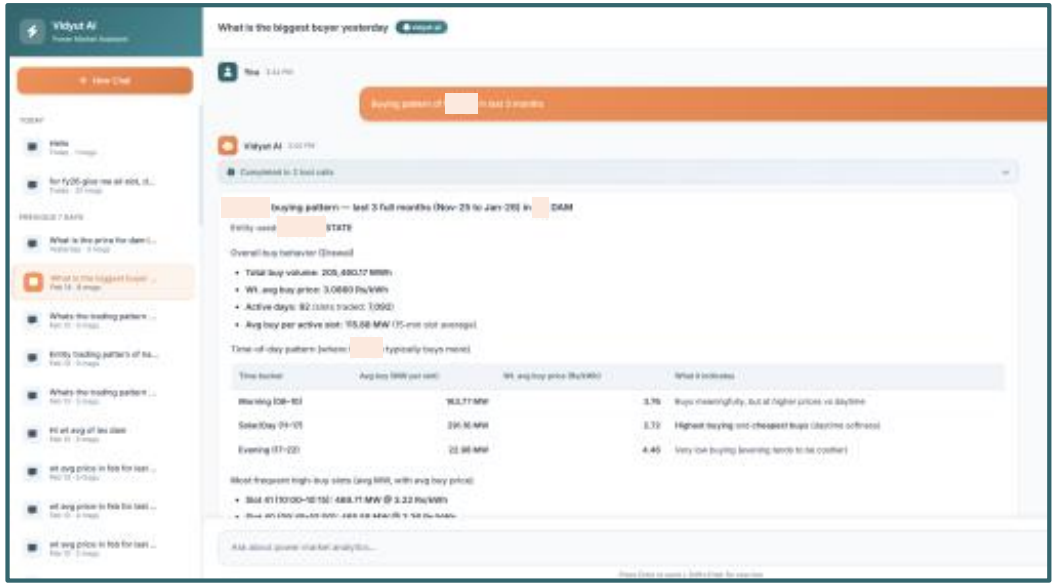
Long-term resource adequacy planning tool



Vidyut Ai – AI Agent for the Indian Power Market

Vidyut Ai (AI Agent) enables users to interact with complex energy data through natural language queries and receive clear, actionable insights anytime and from anywhere.

1. **Natural Language Interaction** – Ask questions and get insights from complex energy data anytime, anywhere.
2. **Rapid Data Retrieval and Summarisation** – Quickly access key information across demand, generation, and procurement.
3. **Tailored Analytical Insights** – Understand system surplus/deficit positions and power procurement outcomes.
4. **Dynamic Report Generation and Interactive Visualisations** – Communicate trends and comparisons effectively.
5. **Supports Operational and Strategic Decision-making** – Provides actionable, data-driven guidance with minimal manual effort.
6. **User-friendly Interface** – Simplifies electricity data analysis, enhancing responsiveness and reducing workload.



GNAi – Key Benefits for the State

- **Access to AI Agent (Vidyut AI):** Access to Indian Power Market's first AI agent
- **Enhanced Operational Efficiency:** Automate routine, time-consuming manual processes
- **Optimisation and Simulation:** Automated modules with simulation capabilities for specific problem statements, such as optimization of power procurement
- **Improved Decision-Making:** Granular operational data combined with AI- powered insights for precise load forecasting, optimal power procurement, and demand-supply analysis
- **Secure and Compliant Data Environment:** Operate with the assurance that all sensitive data is managed with the highest standards of security and privacy of a CERC regulated platform

Key Events and Success Stories

GNA Energy launches Vidyut AI at AI Impact Summit 2026

GNA Energy and GRIDCO's Seminar on Digitizing Odisha's Intra-State Power Market

GNA Energy & Indian Gas Exchange (IGX) Unveil the Gas Analytics Dashboard

GNA Energy at CEA's Brainstorming Conclave on the Indian Power Sector

Media Recognition & Industry Coverage

GNA

GNA Energy is striving to become a key enabler of **New India's Clean Energy Transition**

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